

Michael Xu

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Scholar: [Michael Xu](#) | LinkedIn: [in/yunhuixu](#)

Research engineer focused on LLM post-training and automated red-teaming, with a background in offensive cybersecurity.

EXPERIENCE

Senior Research Engineer

Sept. 2024 – Present

Microsoft Research - Deep Learning Group

San Francisco, CA

- Co-developed GFlowRL, a distribution-matching RL objective that replaces FlowRL's learned log Z network with an in-batch Monte Carlo estimate; outperformed GRPO/PPO on math and code, reaching 2048 Codeforces Elo (96th percentile) with a 14B model ([paper](#)).
- Owned end-to-end scaling of on-policy RL post-training to 96 B200 GPUs and 64k context on Megatron-LM, supporting dense and 235B+ parameter Mixture-of-Experts models with hybrid attention and multi-token prediction.
- Built containerized sandboxes for executing untrusted model-generated code during RL rollouts, converting automated test-suite feedback into verifiable reward signals.
- Extended GFlowRL to multi-turn adversarial red-teaming, achieving 82.5% average ASR@1 on AdvBench across three instruction-tuned victim models, 2.4 points above the prior SOTA attacker.
- First-authored Rushes, a public benchmark of 44,226 revealed preferences from 8,167 Xbox Insiders across six games; showed GPT-5 trailing personalized matrix factorization and a popularity baseline ([paper](#)).

Research Engineer

Feb. 2021 – Sept. 2024

Microsoft Research - Natural Language Processing

San Francisco, CA

- Built *Craft an Iron Sword*, an agentic LLM system for Minecraft combining composable tool calling with dynamic RAG retrieval via JavaScript execution and promise chaining; shipped in 2021, ahead of ReAct and vendor function-calling APIs ([paper](#), 67 citations, [GitHub](#)).
- Added real-time video and voice multimodality; showcased by Satya Nadella at Build 2024 ([video](#)) after winning Executive Hack at Microsoft Hackathon 2022.
- Developed a structured conversational prompting framework to improve LLM alignment, raising quest scenario completion rates by 25% ([paper](#)).

SELECTED PUBLICATIONS

GFlowRL: Scaling Distribution-Matching RL to Large Language Models — Liu*, **Xu***, Stokes, Smolensky, Burger, Gao. arXiv, 2026. *Equal contribution.

Rushes: A Human Preference Dataset for Pluralistic Alignment — **Xu**, Leandro, Rao, et al. arXiv, 2026.

Weak-to-Strong On-Policy Distillation — Yu, W. Xu, **M. Xu**, Zhou, Lin. arXiv, 2026.

EDUCATION

University of Illinois at Urbana-Champaign

Champaign, IL

M.S. in Computer Science, Data Science Concentration - 3.91 GPA

Jan. 2022 – Dec. 2023

Arizona State University

Tempe, AZ

B.S. in Computer Science, Minor in Statistics - 4.0 GPA

Aug. 2017 – Dec. 2020

- Second person to earn [pwn.college's Blue Belt](#) (CSE 466, Offensive Security): binary exploitation, reverse engineering, shellcode, and sandbox escapes.

TECHNICAL EXPERTISE

Research Areas: RL Post-Training (GRPO, PPO, Trajectory Balance), RLVR & RLHF, On-Policy Distillation, LLM Alignment, Automated Red-Teaming & Cybersecurity

ML & Infrastructure: Python, PyTorch, slime, veRL, Megatron-LM, vLLM, Transformers, FSDP / DeepSpeed, Ray, TensorRT, Kubernetes, Docker, Azure